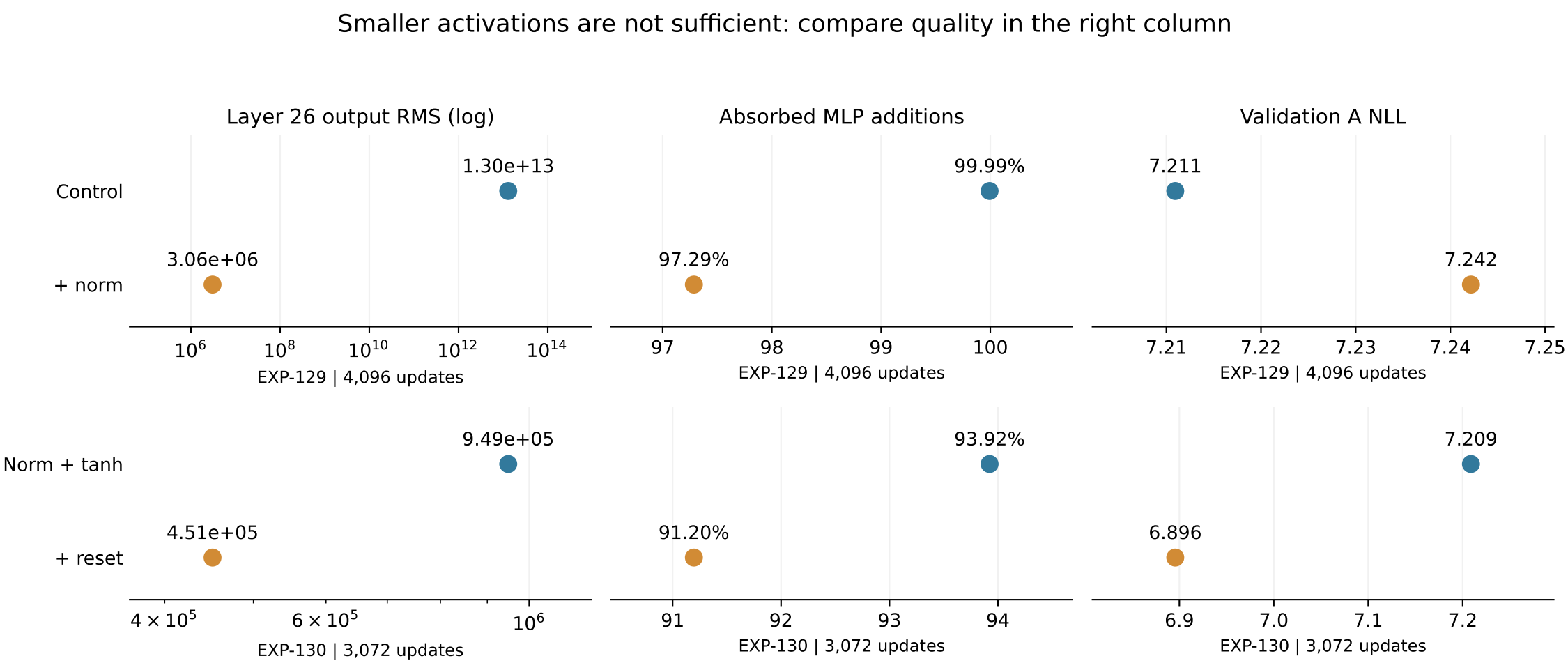
Smaller activations are not sufficient: compare quality in the right column
Layer 26 output RMS (log)
Absorbed MLP additions
Validation A NLL
Control
+ norm
1.30e+13
3.06e+06
99.99%
97.29%
7.211
7.242
10^6
10^8
10^{10}
10^{12}
10^{14}
97
98
99
100
7.21
7.22
7.23
7.24
7.25
EXP-129 | 4,096 updates
EXP-129 | 4,096 updates
EXP-129 | 4,096 updates
Norm + tanh
+ reset
9.49e+05
4.51e+05
93.92%
91.20%
7.209
6.896
4×10^5
6×10^5
10^6
91
92
93
94
6.9
7.0
7.1
7.2
EXP-130 | 3,072 updates
EXP-130 | 3,072 updates
EXP-130 | 3,072 updates
Rows are separate matched comparisons. Layer measurements: one diagnostic window; NLL: all 64 validation windows.
The normalization reduces RMS by millions of times, yet NLL slightly worsens. Reset improves NLL as well.